KMT-2026-BLG-0521 / OGLE-2026-BLG-0058

Astrometry of the microlensing target — summary

Prepared 2026-09-10. Solution: `~/KMTdata/Results/v6/IFfinal_<field>.mat` (variable `IFsys`), tie in `Tie_<field>.mat`.

1. The data and the algorithm

Data. Two KMTNet cut-outs of the same field, **BLG41** and **BLG01**, 300 x 300 pixels at 0.4 arcsec/pix, in I band, spanning 2016 to 2026 in ten observing seasons (2020 is a short stub and is dropped). **19 133** and **19 981** exposures respectively, taken alternately on the same nights, a median of 12.2 minutes apart. The target sits at the field centre at I_{OGLE} = 18.13, RA 17:52:38.09, Dec –31:47:36.1.

Calibration set. The analysis is restricted to stars selected on magnitude and isolation alone: **14 < I < 19** with **no companion brighter than I = 18 within 2 arcsec**, the companion list taken from the OGLE catalogue, which resolves about six times more sources than KMT detects. This leaves **287** stars on BLG41 and **253** on BLG01. The target passes the selection on its own merits. Everything else is dropped at the input, so all measurement, detrending and quality cuts act on this set alone.

The fit. Each exposure gets a full 2-D affine transformation, 6 parameters, and each star a position and proper motion, 4 parameters; these are solved together by alternating least squares. Alongside them the fit removes differential chromatic refraction (fitted in 6 equal-population colour bins), an annual term, a pixel-phase term, and two SysRem components computed once over the whole decade. Every star is freely fitted — nothing is held at a catalogue value — and each is weighted by its own residual scatter, with moving-median outlier rejection.

Absolute frame. A free fit determines only *relative* astrometry, so the result is tied to Gaia DR3 afterwards. The tie is fitted on stars with **RUWE < 1.4 and 15 < I < 17** and removes the full six-parameter gauge — a constant *and* a linear gradient across the field — then is applied to every source.

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2. Results

Achieved accuracy, residual RMS over the decade after position and proper motion are removed:

	BLG41 ΔX / ΔY	BLG01 ΔX / ΔY
calibration stars brighter than I = 17	**5.15 / 5.06 mas**	**4.93 / 5.12 mas**
the target, I = 18.13	**23.75 / 19.63 mas**	**21.36 / 20.37 mas**

The frame is therefore good to about **5 mas** on well-measured stars, and the target — three magnitudes fainter and blended — is measured to about **20 mas** per epoch, consistent with the running median of the residual RMS at its magnitude.

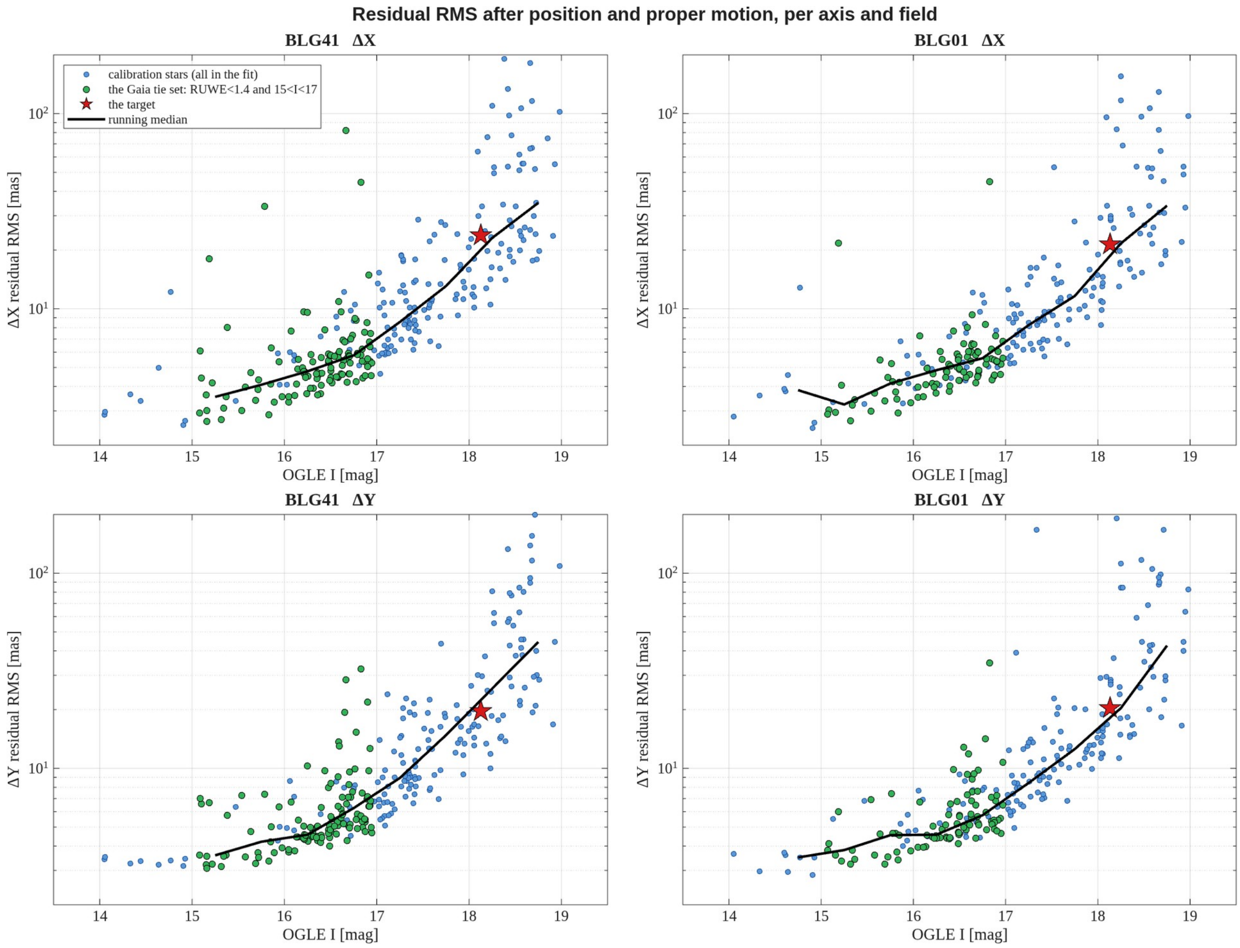
Proper motion of the target, absolute, after the Gaia tie:

	$\mu_\alpha \cos\delta$	μ_δ
BLG41	**–1.937 ± 0.335**	**–7.140 ± 0.360**
BLG01	**–1.658 ± 0.256**	**–7.089 ± 0.377**

difference between the fields	−0.279 ± 0.421	−0.052 ± 0.521

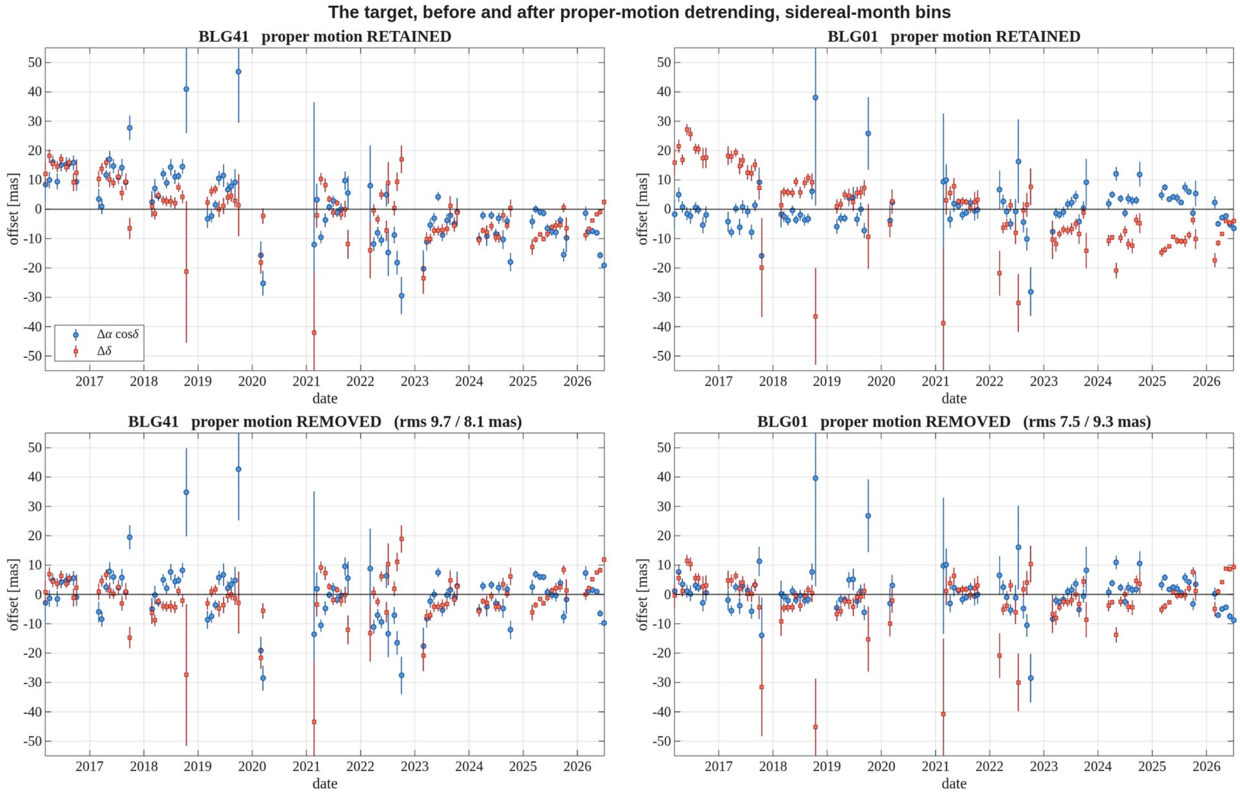
mas/yr. The two independent fields agree within their errors, at 0.66 and 0.10 sigma. The quoted error is internal, from the scatter of the ten season means about the fitted line; the external check against Gaia gives about **1.0 / 0.65 mas/yr** per star, and that is the figure to use in practice.

No astrometric anomaly is detected at the target. Its χ^2 on monthly binned residuals is 1574 and 1313 for 180 degrees of freedom, against field medians of the same quantity — the 63rd and 53rd percentile of the population, and the 68th and 62nd among stars within 0.4 mag of it. Whatever astrometric excursion the microlensing event produces is smaller than the scatter of an ordinary star of this brightness.



Residual RMS against the OGLE catalogue magnitude, per axis and per field, after position and proper motion have been removed. Blue: all 287 / 253 calibration stars, every one freely fitted. Green: the stars the Gaia tie is fitted on, RUWE below 1.4 and 15 < I < 17. Red: the target. Black: running median. The dashed lines mark the 14 and 19 mag selection limits.

`report_v6_RMS.png`



The target before and after proper-motion detrending, in both fields. Columns are the fields; the top row keeps the proper motion and the bottom row removes it. Blue circles are the offset in right ascension, red squares in declination, binned in one sidereal month with the standard error within the bin. The proper motion is plainly visible as the declination trend in the top row and is absent from the bottom row; the residual RMS quoted in the lower panels is per bin.

`report_v6_target_beforeafter.png`

More details

3. Object selection, step by step

step	BLG41	BLG01
raw sources in the MSc file	1177	1187
survive the standard quality cuts	594	621
have an OGLE I magnitude	540	478
in 14 < I < 19	529	470
... and isolated: no I < 18 companion within 2.0 arcsec	**287**	**253**

The isolation cut is the dominant one, removing 46% of the magnitude-selected sources on both fields. For the tie only, Gaia is required: 273 of 287 and 225 of 253 sources have a counterpart, of which 220 and 177 pass RUWE < 1.4, and 118 and 90 also fall in 15 < I < 17.

4. The Gaia tie

The map. Gaia standard coordinates are fitted directly onto cut-out pixels from the tie stars — not carried through OGLE, which would inherit its ~110 mas astrometric error. Median residual **23.4 mas (BLG41)** and **22.1 (BLG01)**, with scales of 2.5195 and 2.5290 pix/arcsec against the 2.500 expected at 0.4″/pix. Restricting the tie set to 15 < I < 17 improves the map by 35 to 40% over using all RUWE-clean stars, since bright stars have better positions.

The gauge. In the proper motions the gauge freedom is a constant plus a linear gradient across the field — six parameters. Removing only the constant leaves a shear of **−0.037 mas/yr per pixel**, visible as `corr(Δμ_α, Y) = −0.75` and reproduced at nearly the same value in both independently reduced fields. Removing the full gauge:

scatter of our proper motions about Gaia	constant only	full gauge
BLG41	4.34 / 3.57 mas/yr	**1.02** / **0.64**
BLG01	3.48 / 3.20 mas/yr	**1.06** / **0.67**

What the magnitude cut on the tie set is worth. Evaluated on the same test set — all RUWE-clean stars, so neither tie is flattered by its own training sample — it changes the proper-motion agreement by a few per cent, inconsistent in sign. The narrower set is better measured per star, 0.57 against 0.98 mas/yr, and determines the gauge slightly more precisely, 0.053 against 0.066 mas/yr, but halving the star count nearly cancels that and the gauge is **under 1% of the total variance**. The cut is kept because it improves the positional map, not because it improves the motions.

5. What the fit does, in detail

Each of the two steps runs **two passes** internally. `IFsys.AnnualEffect` and `IFsys.PixPhase` read `0` in the saved solutions, which does *not* mean those effects were ignored: they are fitted and subtracted from the data in pass 1 and switched off in pass 2, which works on data from which they have been removed.

	pass 1	between	pass 2 (final)
per-epoch affine	fitted		fitted
per-source position and motion	fitted		fitted
DCR / chromatic refraction	fitted		fitted again
annual term	fitted and subtracted		forced off
pixel-phase term	fitted and subtracted		forced off
SysRem		applied	
iterative reweighting	15 iterations		6 iterations

- **Per-epoch affine** — 6 parameters per exposure in `ParE`, stored as the deviation from the identity: translation (2), rotation (1), uniform scale (1), shear (2). 574 equations for 6 unknowns each epoch. Translation dominates at about 37 / 28 mas rms; rotation and scale each move a star at the cut-out edge by roughly 10 mas.
- **Per source** — position at `JD0` = 2019-06-01 and proper motion, 4 parameters in `ParS`, in pixels and pixels per year at 400 mas/pix.
- **DCR** — `[1, sin(pa)·secz, cos(pa)·secz]` plus six higher orders in `(secz·sin pa)^n`, `(secz·cos pa)^n` for n = 2, 3, 4: 18 parameters per colour bin, 6 equal-population bins. Only the *differential* part is recoverable — refraction common to the field is exactly a translation and is absorbed by `t_x, t_y` every epoch.
- **Annual term** — a quartic in the phase of the calendar year, 10 parameters, fitted globally, one curve shared by every star. Not parallax, which is per-source and switched off.
- **Pixel phase** — a quintic in the sub-pixel phase, 10 parameters, global, recoverable only because the pipeline stored the per-epoch registration shifts.
- **SysRem** — 2 components over the whole decade, reused rather than refitted per season, with point-wise rejection of bad cells.

Timings and convergence:

	BLG41	BLG01
step 1, decade SysRem	19.8 min	16.7 min
step 2, final solution	35.6 min	31.3 min
convergence, last 3 iterations	0.0028 mas	0.0007 mas

No per-season fitting enters the solution; seasons are used only for reporting.

6. Comparison stars

Six, chosen to be the same physical objects in both fields — three near the target's magnitude and three brighter:

tag	I	dist [pix]	decade RMS ΔX / ΔY, BLG41	BLG01
`m18_d04`	18.14	4.5	29.03 / 27.91	24.84 / 26.86
`m18_d17`	18.09	17.3	51.68 / 27.85	85.47 / 28.47
`m18_d48`	18.05	48.4	17.92 / 16.76	14.58 / 15.57
`m16_d24`	16.68	24.4	**4.19** / **4.27**	**4.17** / **4.38**
`m16_d30`	16.63	30.4	6.91 / 5.85	6.43 / 5.55
`m16_d31`	16.75	30.6	8.48 / 7.79	11.20 / 7.71

mas. The bright trio reproduces between the fields almost exactly — `m16_d24` gives –3.31 / –7.81 on BLG41 and –3.31 / –7.89 on BLG01 — while the faint trio does not: `m18_d17` gives –6.03 against –10.05 mas/yr and reaches 52 and 85 mas decade RMS on only ~2400 usable epochs. It is a poor star.

`m18_d04` is the neighbour blended with the target, 4.5 pixels away under ~7 pixel seeing. It is a legitimate member of the calibration set — the target at I = 18.13 is fainter than the I < 18 companion threshold, so it does not trigger rejection — but its centroid tracks the target's brightening through the event, and it is not an independent object.

Coordinates. Pixel positions are per field, since the cut-outs are offset by about 1.1 pixels; sky positions come from inverting each field's own map and the two fields agree on them to 2 to 9 mas.

star	I	BLG41 pixel	BLG01 pixel	RA (J2000)	Dec (J2000)
target	18.13	150.4, 150.4	151.6, 150.2	17:52:38.085	–31:47:36.21
`m18_d04`	18.14	146.5, 148.3	147.6, 148.0	17:52:38.210	–31:47:37.06
`m18_d17`	18.09	134.5, 143.6	135.6, 143.4	17:52:38.590	–31:47:39.00
`m18_d48`	18.05	198.6, 145.4	199.8, 145.0	17:52:36.585	–31:47:38.10
`m16_d24`	16.68	161.3, 172.2	162.4, 172.1	17:52:37.746	–31:47:27.53
`m16_d30`	16.63	120.4, 154.6	121.4, 154.5	17:52:39.023	–31:47:34.60
`m16_d31`	16.75	138.7, 178.7	139.8, 178.6	17:52:38.436	–31:47:24.98

7. Chi-squared of the monthly binned residuals

For every star the residuals are binned in sidereal months; each bin contributes `(mean/standard error)²` in both axes, and `DoF = 2·N_bins − 4`. Values are not divided by DoF.

	BLG41	BLG01
stars measured	287	253
median DoF	182	182
target χ^2	**1574** (DoF 180)	**1313** (DoF 180)
target χ^2/DoF	8.7	7.3
field median χ^2/DoF	7.2	7.1
target percentile, all stars	63%	53%
target percentile, stars within 0.4 mag	68%	62%

The target is unremarkable. The field median of χ^2/DoF ≈ 7 is itself informative: the standard error of the mean within a month **understates the true bin uncertainty by about $\sqrt{7} \approx 2.6$**, because the noise within a month is correlated rather than independent. The effect is multiplicative and nearly independent of magnitude — the Spearman correlation of χ^2/DoF with I is −0.12 and −0.17 — and the faintest stars come closest to unity, since for them genuine photon noise, which *is* independent between exposures, dominates.

8. Overlap of the two fields

Stars are identified through the OGLE catalogue; matching the two source lists directly is ambiguous because a close pair may be split in one field and merged in the other.

	BLG41	BLG01
sources	594	621
matched to an OGLE entry	551 (93%)	497 (80%)
distinct OGLE stars behind them	**454**	**394**

The gap between the last two rows is the blending, measured: 551 detections correspond to only 454 stars on BLG41, so about 97 detections are duplicates.

OGLE I	BLG41	BLG01	in both	distinct	found in both
< 15	28	22	22	28	78.6%
15 – 16	54	42	42	54	77.8%
16 – 17	172	149	146	175	83.4%
17 – 18	135	119	117	137	85.4%
18 – 19	55	44	41	58	70.7%
all brighter than 19	**444**	**376**	**368**	**452**	**81.4%**

About four stars in five are detected in both fields, flat from the brightest down to I = 18. **There is nothing fainter than I**

= 19 to compare: our source lists span 12.87 to 18.98 in both fields, while OGLE itself reaches 21.40 and holds 2660 entries fainter than 19. That is the KMT detection limit in this crowding, not a catalogue cut.

9. Files

file	content
`report_v6_RMS.png`	residual RMS against OGLE I, per axis and field
`report_v6_target_beforeafter.png`	the target before and after proper-motion detrending
`report_v6_pm_vs_gaia.png`	our absolute proper motion against Gaia's, after the tie
`report_v6_chi2.png`	χ^2 distribution of the monthly binned residuals, target marked
`report_v6_radec_<field>_<tag>.png`	RA and Dec residuals against time and against each other, colour-coded by time
`report_v6_motion_<field>_<tag>.png`	2x2 motion figures, proper motion retained and removed
`source_motion_v6_<field>[_<tag>].csv`	per-epoch positions for the target and each comparison star

CSV columns are `JD, X_mas, Y_mas, errX_decade, errY_decade, errX_season, errY_season, season`, with positions relative to that star's own mean and **proper motion not removed**. Headers carry the star's magnitude, pixel position, sky position and absolute proper motion.

In the motion figures the columns are the two axes, **X left and Y right**; the top row keeps the proper motion and the bottom row removes it. The legend quotes the motion along the pixel axis and the heading quotes it on the sky. **Pixel X runs opposite to RA**, so those two carry opposite signs in X.

Figures

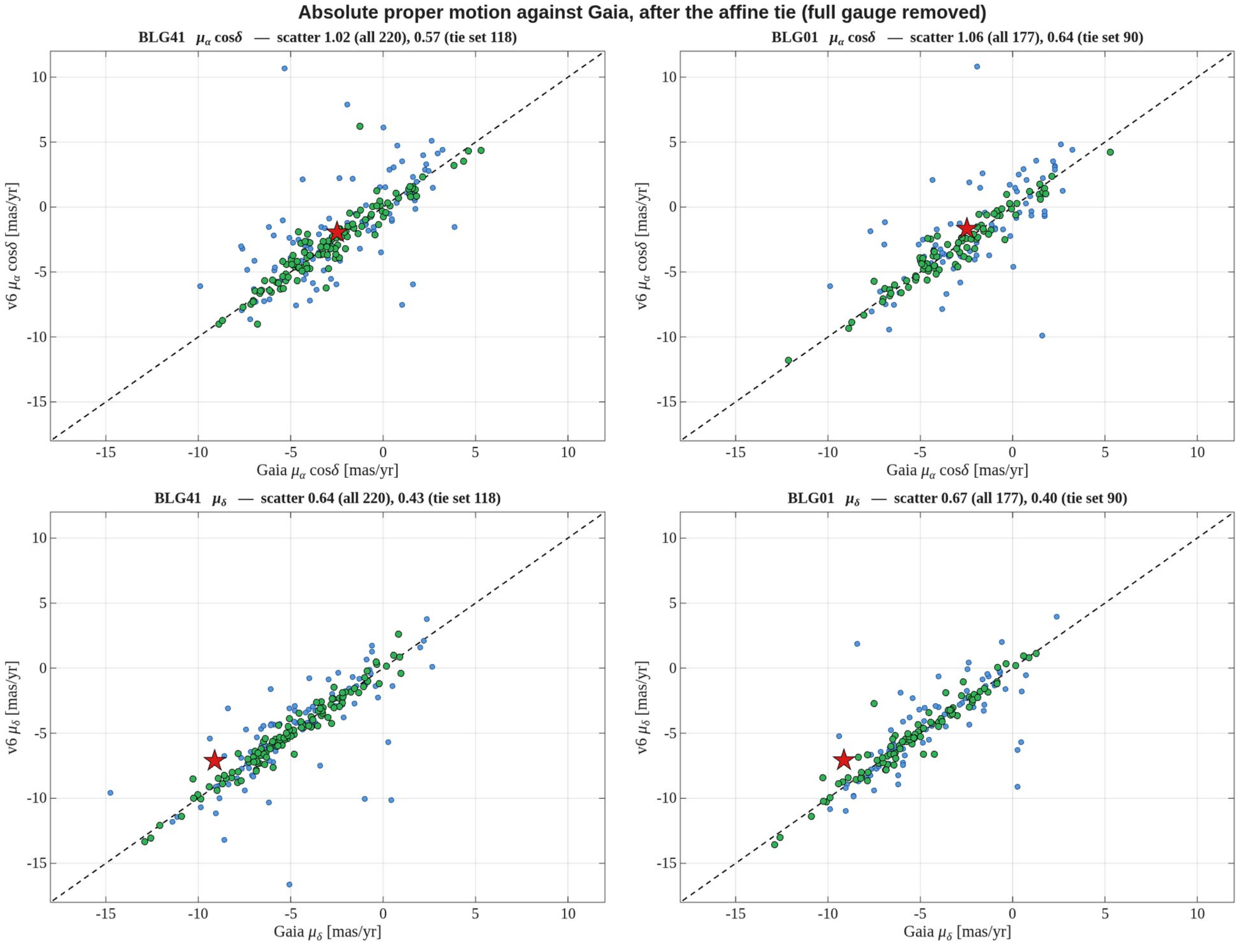


Figure 1. Our absolute proper motion against Gaia's, per axis and per field, after the affine tie with the **full six-parameter gauge** removed. Dashed line is 1:1. The scatter quoted in each panel is the robust standard deviation of the difference; it improves by a factor of three to four over a tie that removes only the constant. The target is the red star.

`report_v6_pm_vs_gaia.png`

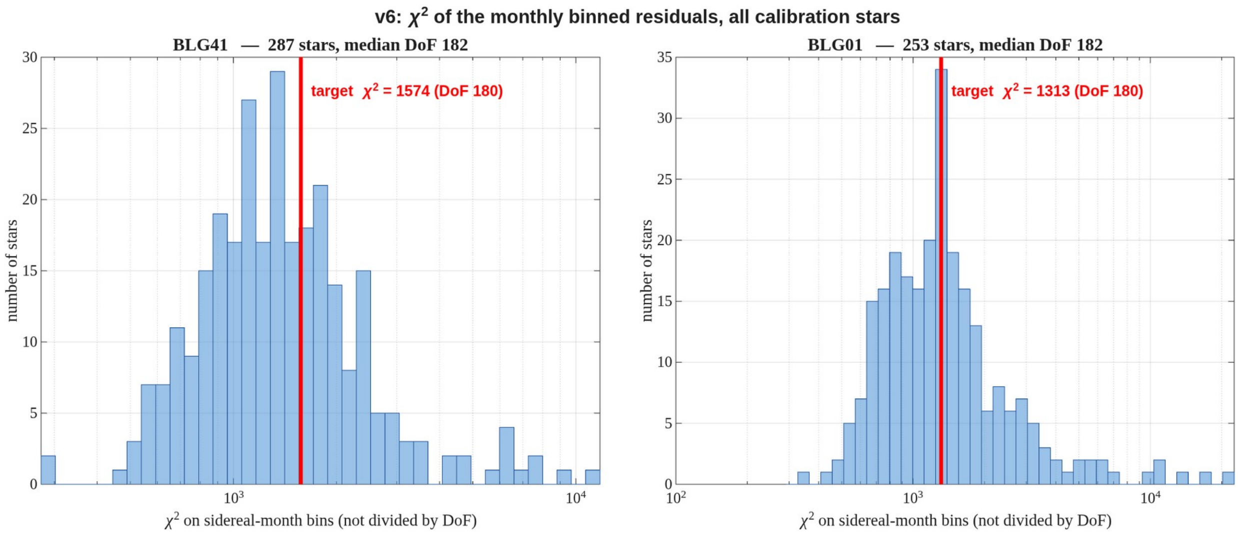


Figure 2. χ^2 of the sidereal-month binned residuals for every calibration star, both axes combined, **not divided by DoF**. Each bin contributes (mean / standard error)² per axis and DoF = $2N_{bins} - 4$. The red line marks the target, which sits in the body of the distribution rather than the tail.

`report_v6_chi2.png`

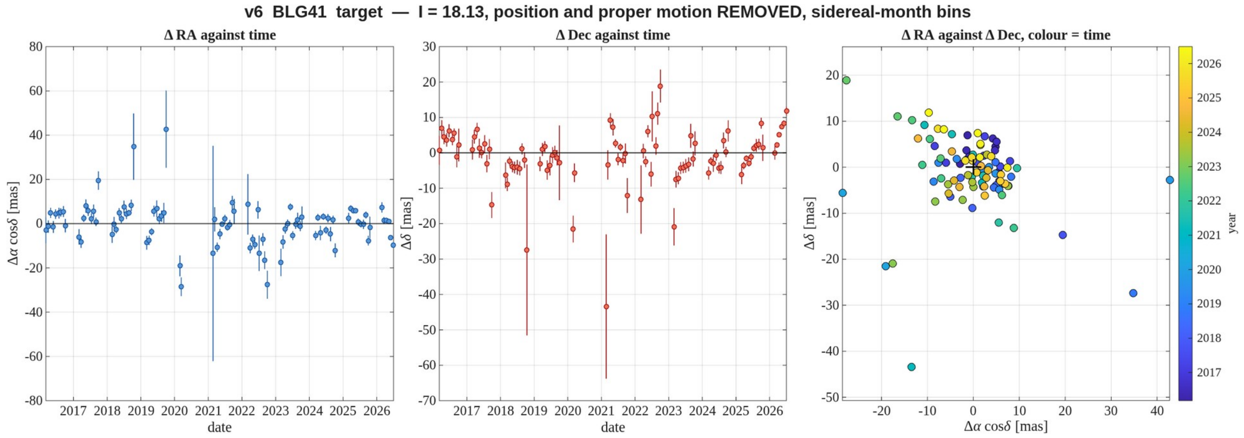


Figure 3. **BLG41** — the target, I = 18.13. **Position and proper motion removed**: these are the residuals. Left: RA residual against time. Middle: Dec residual against time. Right: the two against each other with time colour-coded, where a well-behaved star shows an unordered cloud about the origin and any systematic drift would appear as an ordered progression of colour. Sidereal-month bins; error bars are the standard error within the bin, which section 8 shows understates the true scatter by about 2.6.
`report_v6_radec_BLG41_target.png`

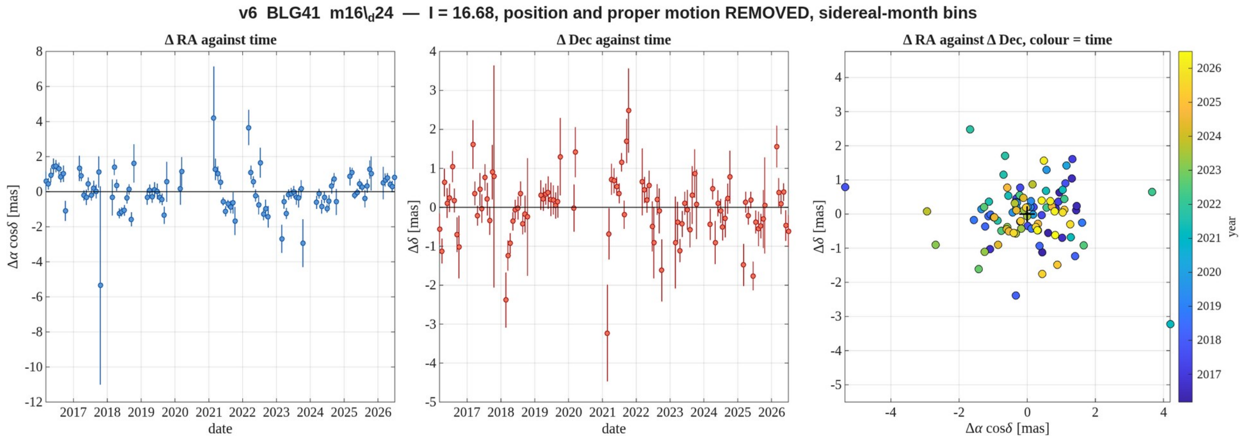


Figure 4. **BLG41** — comparison star I = 16.68, 24 pix from the target — the best measured of the set. **Position and proper motion removed**: these are the residuals. Left: RA residual against time. Middle: Dec residual against time. Right: the two against each other with time colour-coded, where a well-behaved star shows an unordered cloud about the origin and any systematic drift would appear as an ordered progression of colour. Sidereal-month bins; error bars are the standard error within the bin, which section 8 shows understates the true scatter by about 2.6.
`report_v6_radec_BLG41_m16_d24.png`

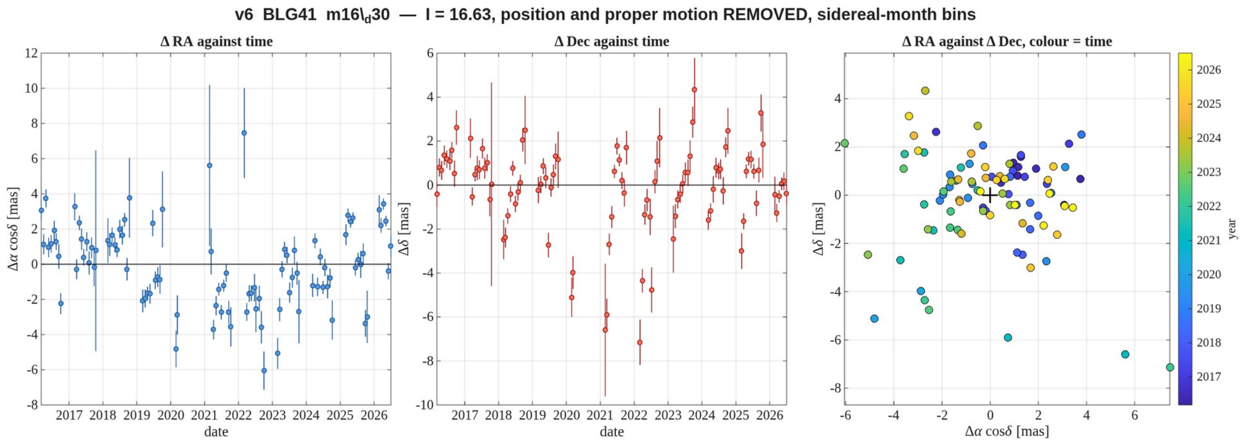


Figure 5. **BLG41** — comparison star I = 16.63, 30 pix from the target. **Position and proper motion removed**: these are the residuals. Left: RA residual against time. Middle: Dec residual against time. Right: the two against each other with time colour-coded, where a well-behaved star shows an unordered cloud about the origin and any systematic drift would appear as an ordered progression of colour. Sidereal-month bins; error bars are the standard error within the bin, which section 8 shows understates the true scatter by about 2.6.

`report_v6_radec_BLG41_m16_d30.png`

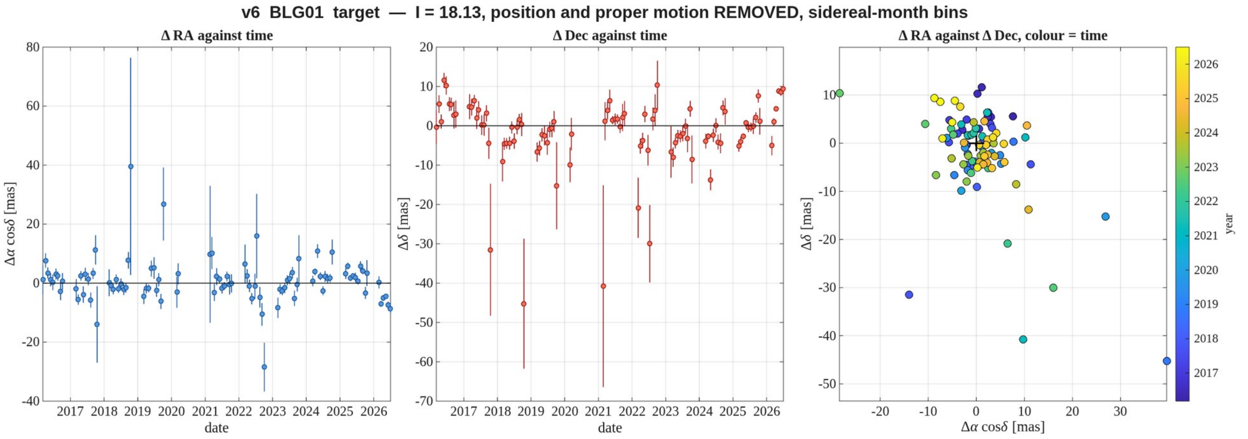


Figure 6. **BLG01** — the target, I = 18.13. **Position and proper motion removed**: these are the residuals. Left: RA residual against time. Middle: Dec residual against time. Right: the two against each other with time colour-coded, where a well-behaved star shows an unordered cloud about the origin and any systematic drift would appear as an ordered progression of colour. Sidereal-month bins; error bars are the standard error within the bin, which section 8 shows understates the true scatter by about 2.6.

`report_v6_radec_BLG01_target.png`

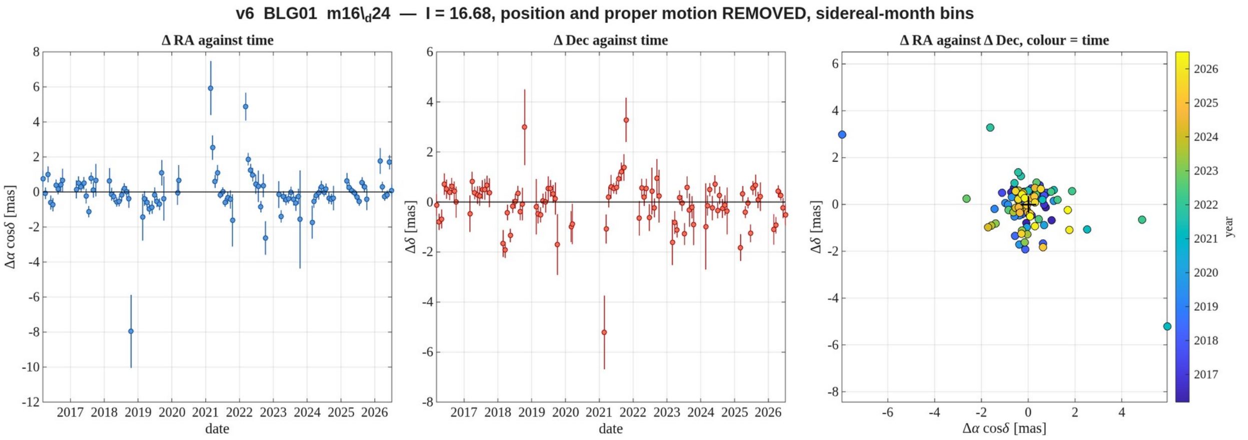


Figure 7. **BLG01** — comparison star I = 16.68, 24 pix from the target — the best measured of the set. **Position and proper motion removed**: these are the residuals. Left: RA residual against time. Middle: Dec residual against time. Right: the two against each other with time colour-coded, where a well-behaved star shows an unordered cloud about the origin and any systematic drift would appear as an ordered progression of colour. Sidereal-month bins; error bars are the standard error within the bin, which section 8 shows understates the true scatter by about 2.6.

`report_v6_radec_BLG01_m16_d24.png`

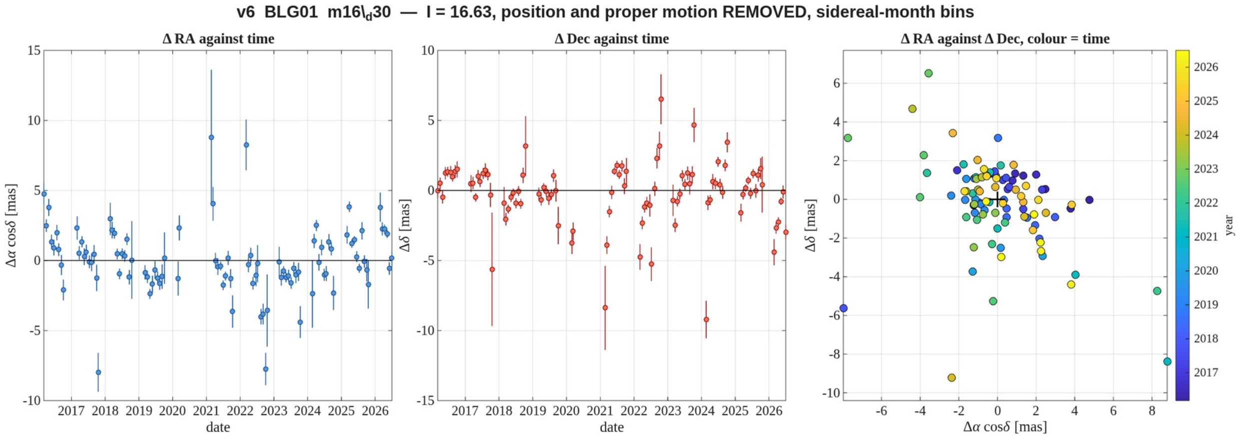


Figure 8. **BLG01** — comparison star I = 16.63, 30 pix from the target. **Position and proper motion removed**: these are the residuals. Left: RA residual against time. Middle: Dec residual against time. Right: the two against each other with time colour-coded, where a well-behaved star shows an unordered cloud about the origin and any systematic drift would appear as an ordered progression of colour. Sidereal-month bins; error bars are the standard error within the bin, which section 8 shows understates the true scatter by about 2.6.

`report_v6_radec_BLG01_m16_d30.png`

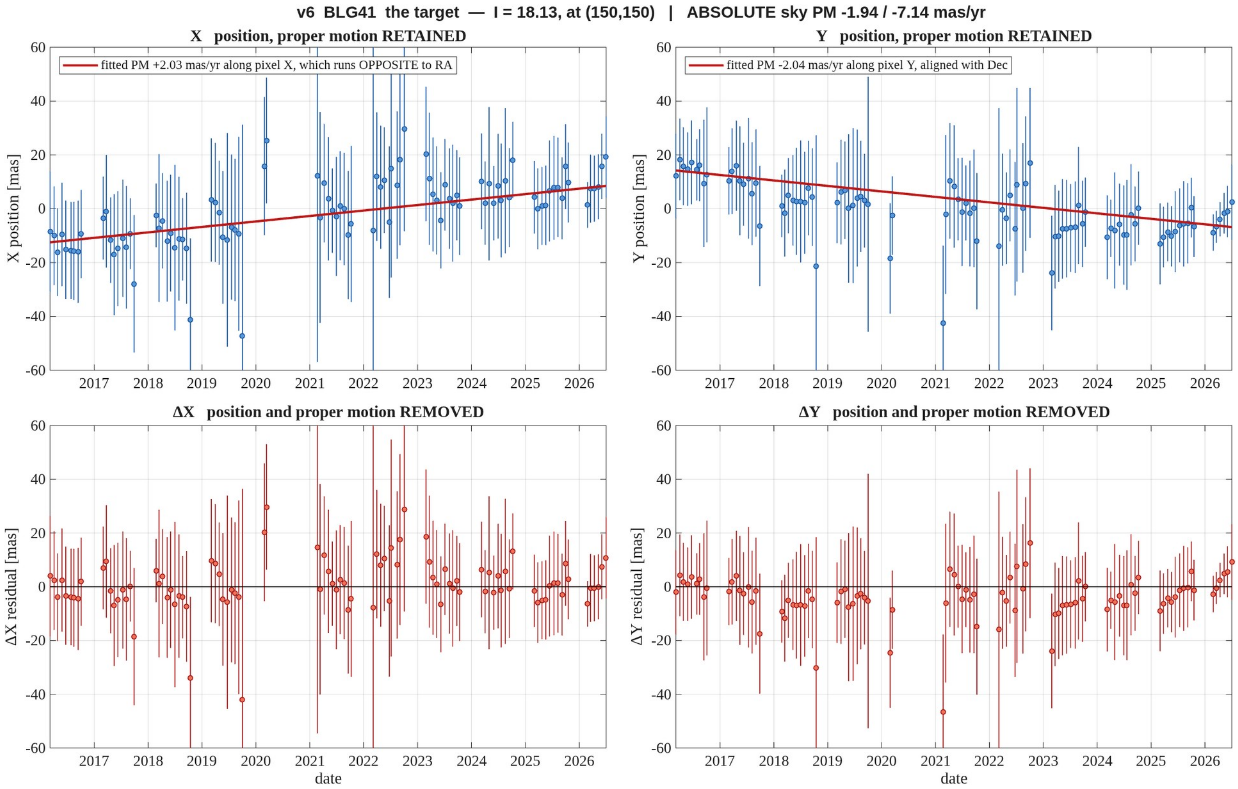


Figure 9. **BLG41** — the target, I = 18.13. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG41_target.png`

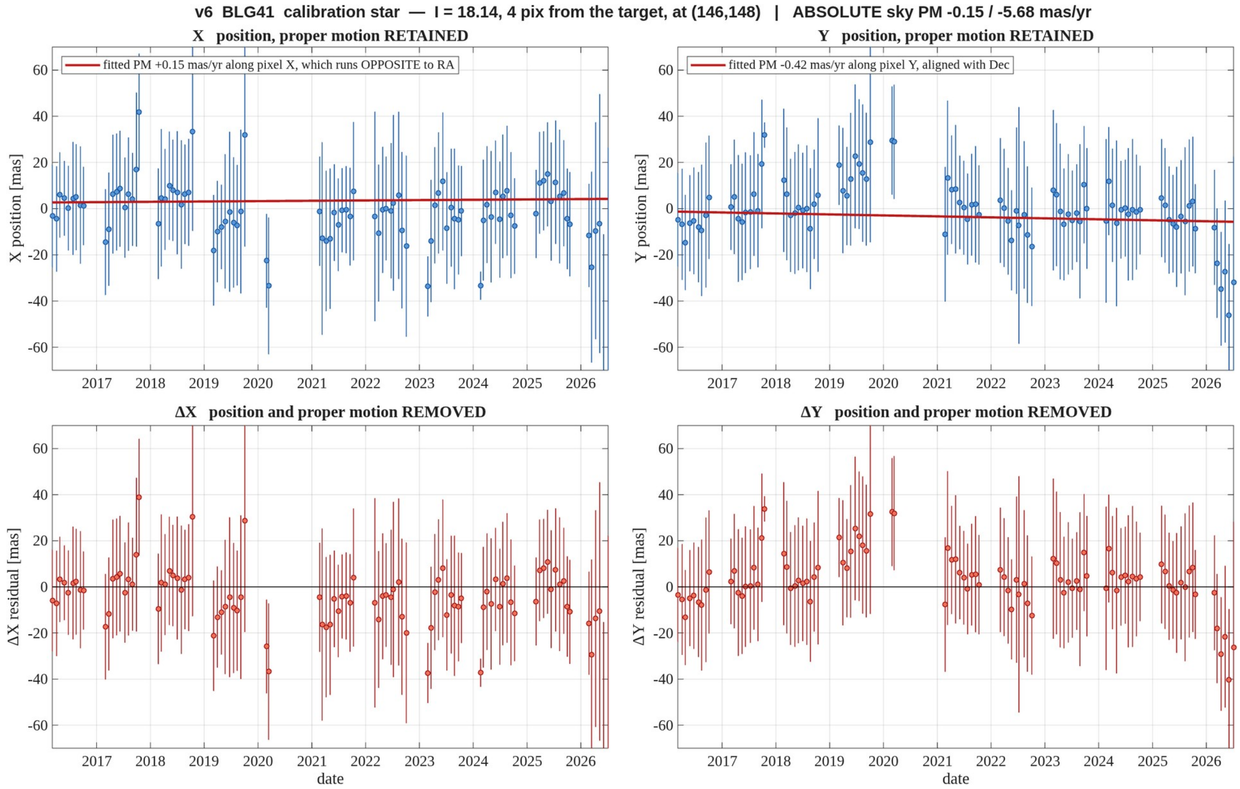


Figure 10. **BLG41** — I = 18.14, 4.5 pix away — **blended with the target**, see section 7. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG41_m18_d04.png`

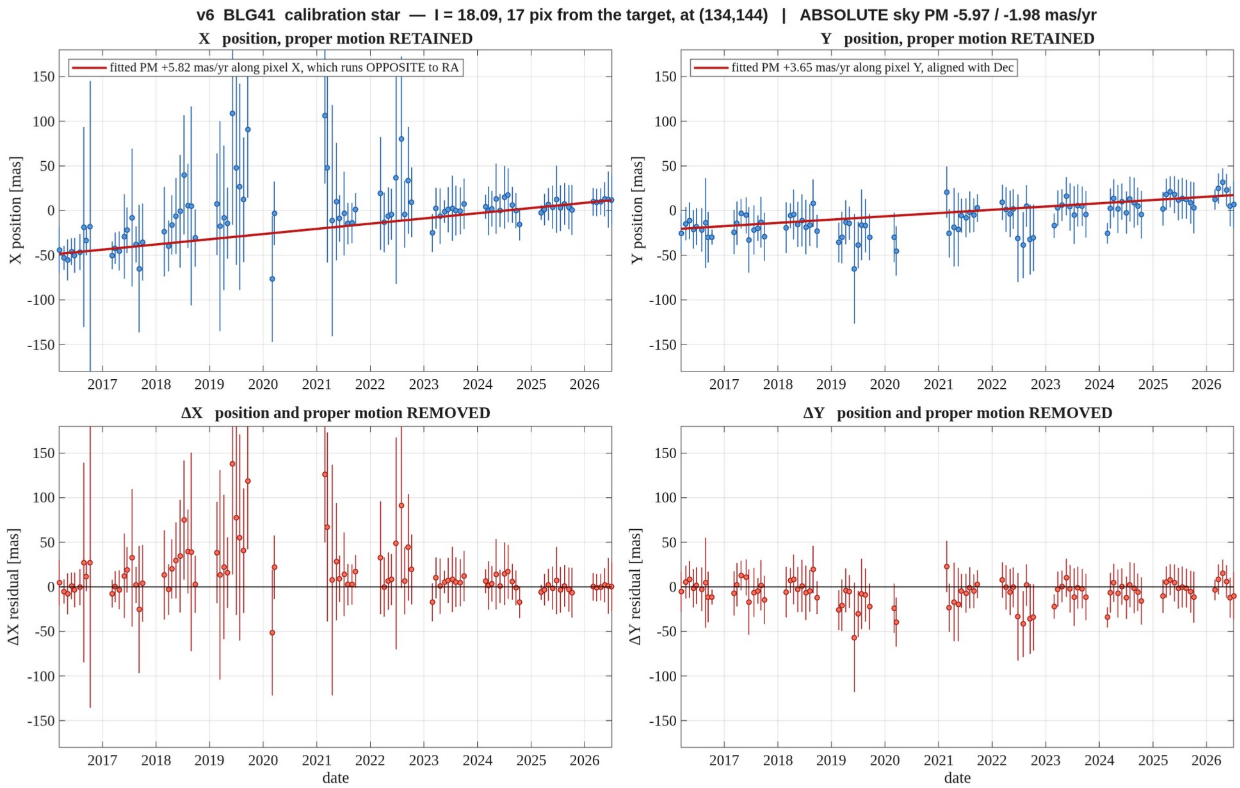


Figure 11. **BLG41** — I = 18.09, 17 pix away — poorly measured, see section 7. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG41_m18_d17.png`

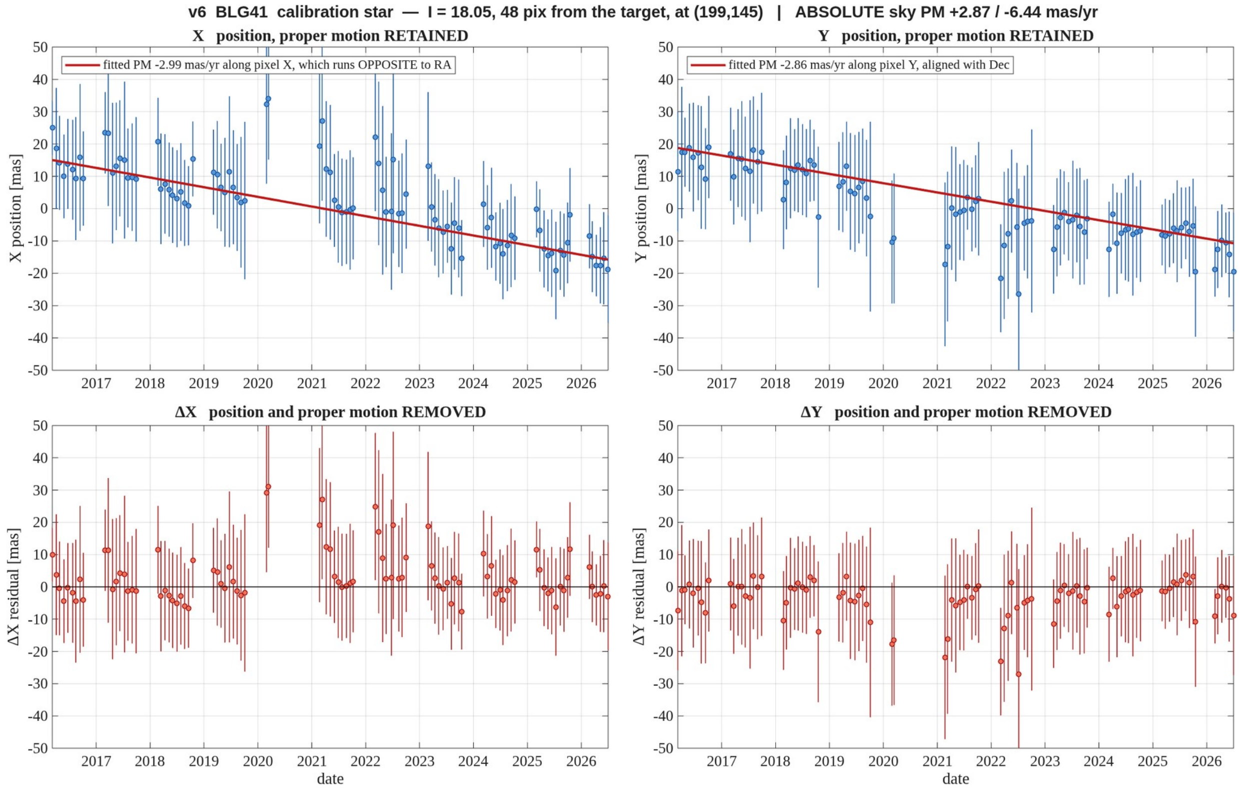


Figure 12. **BLG41** — I = 18.05, 48 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG41_m18_d48.png`

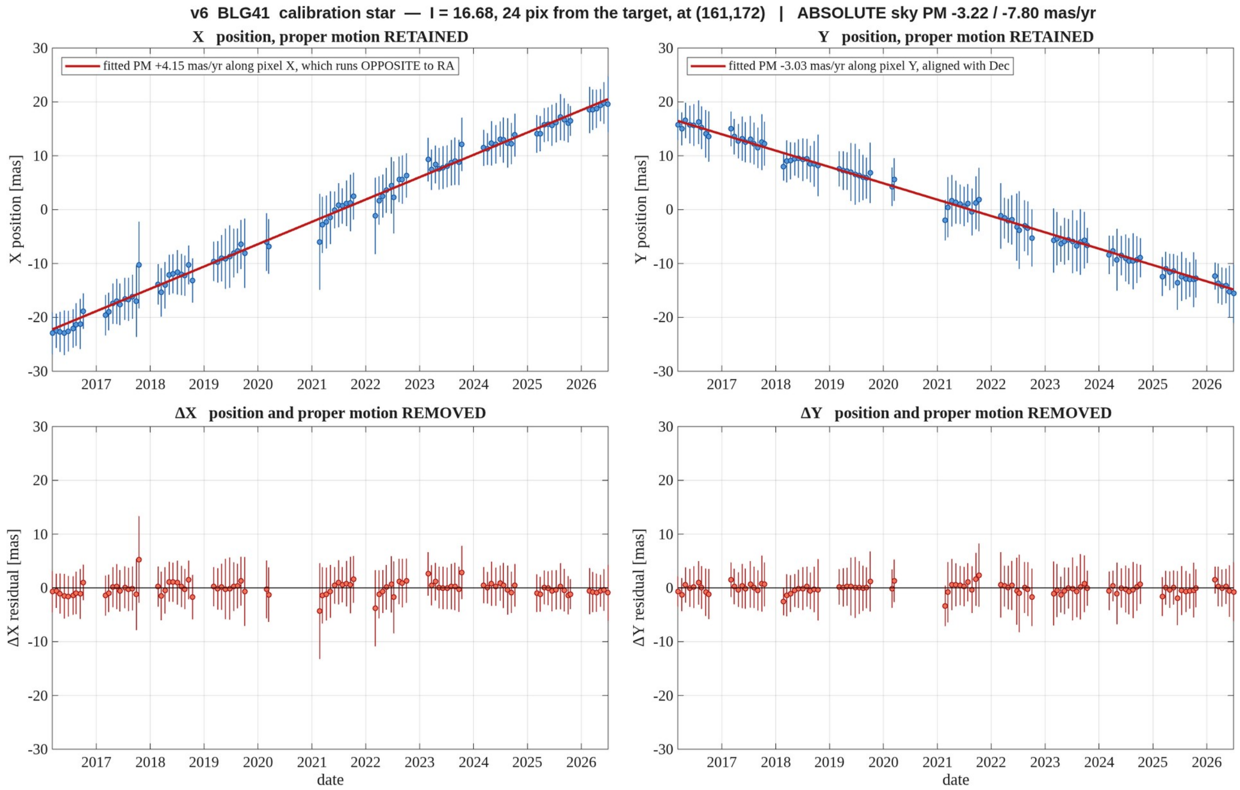


Figure 13. **BLG41** — I = 16.68, 24 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG41_m16_d24.png`

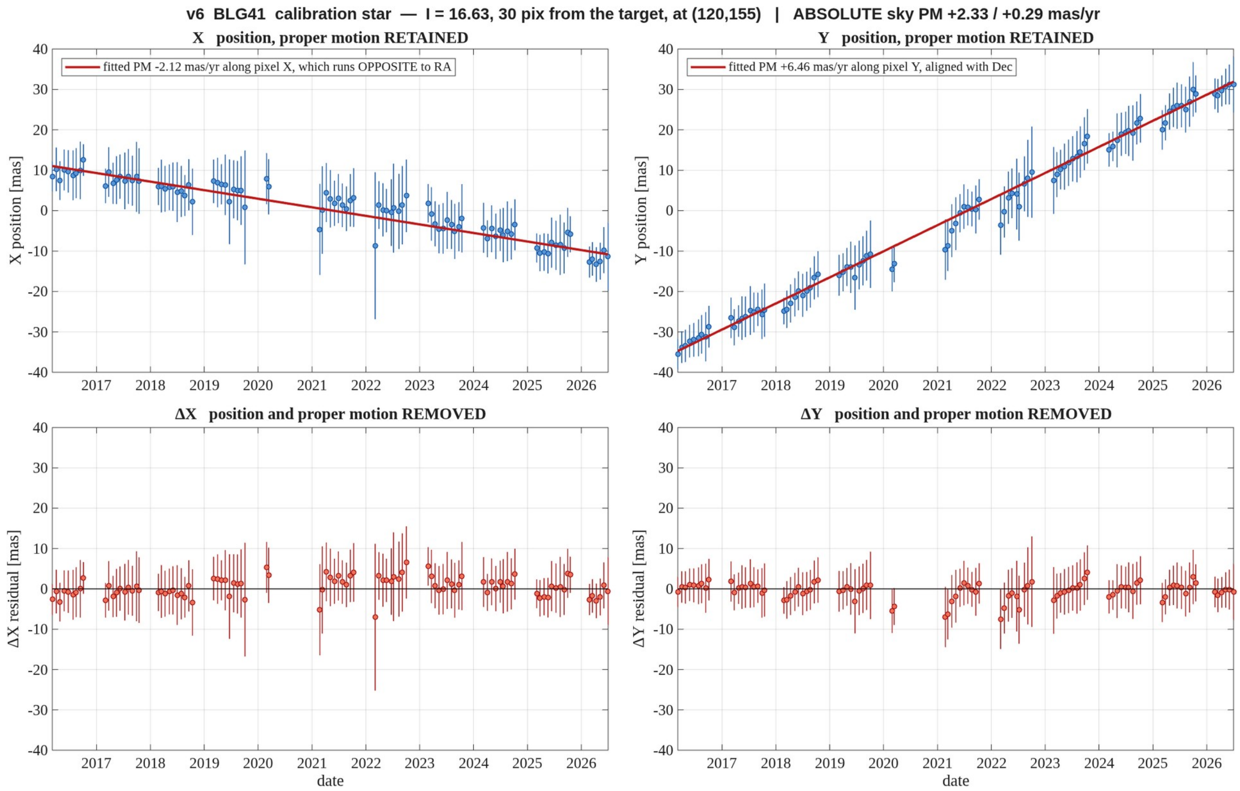


Figure 14. **BLG41** — I = 16.63, 30 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG41_m16_d30.png`

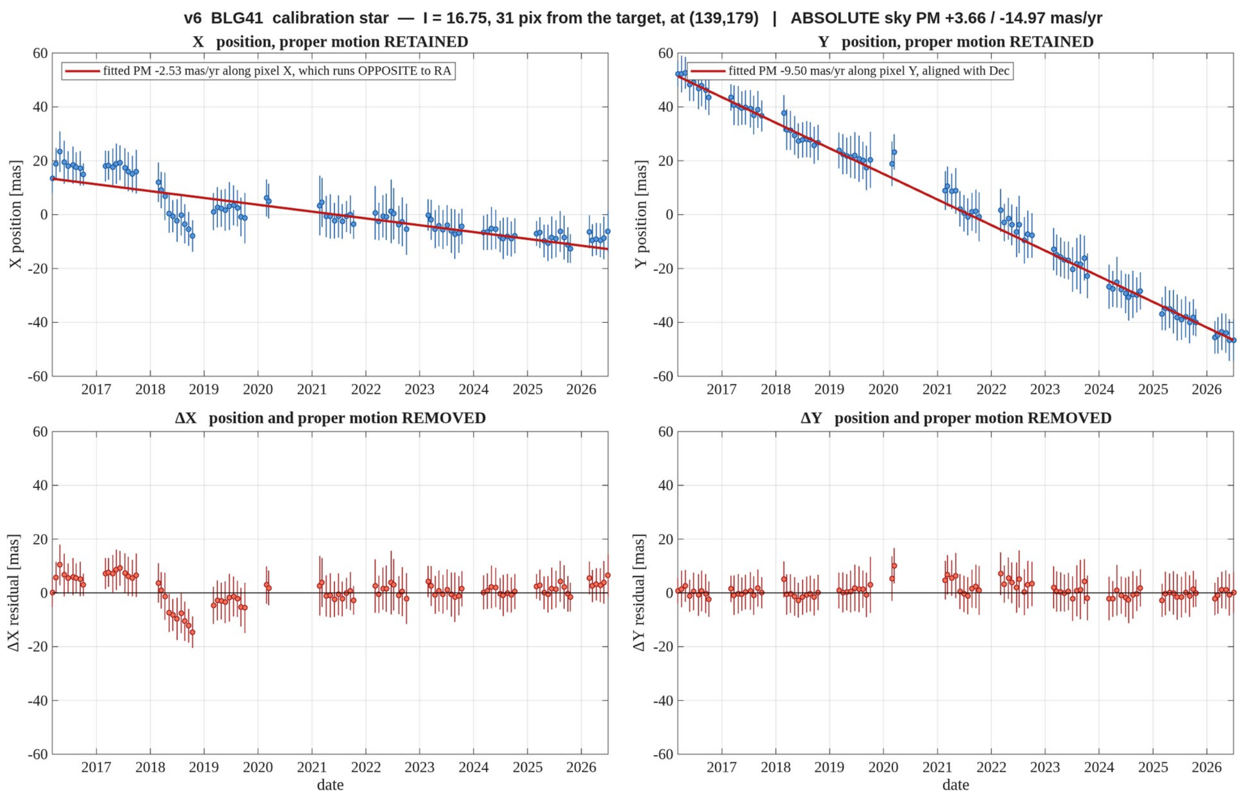


Figure 15. **BLG41** — I = 16.75, 31 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG41_m16_d31.png`

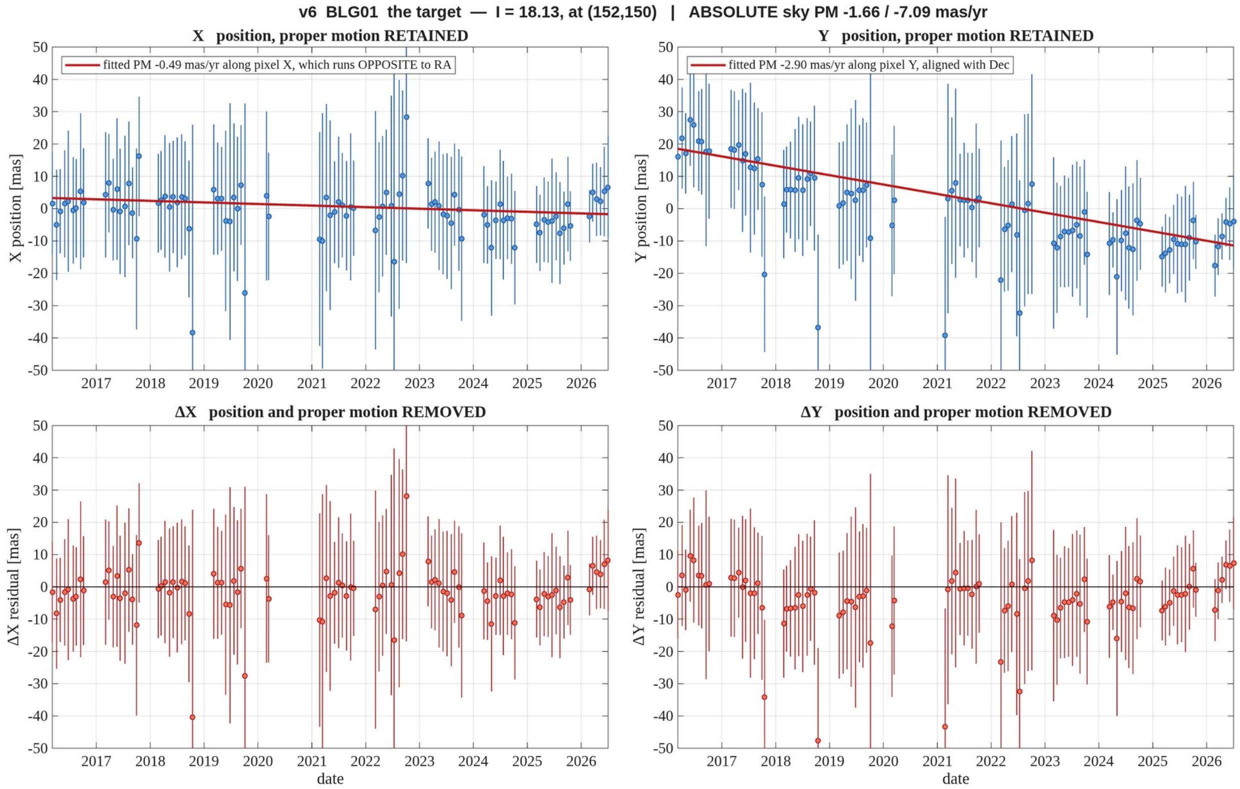


Figure 16. **BLG01** — the target, I = 18.13. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG01_target.png`

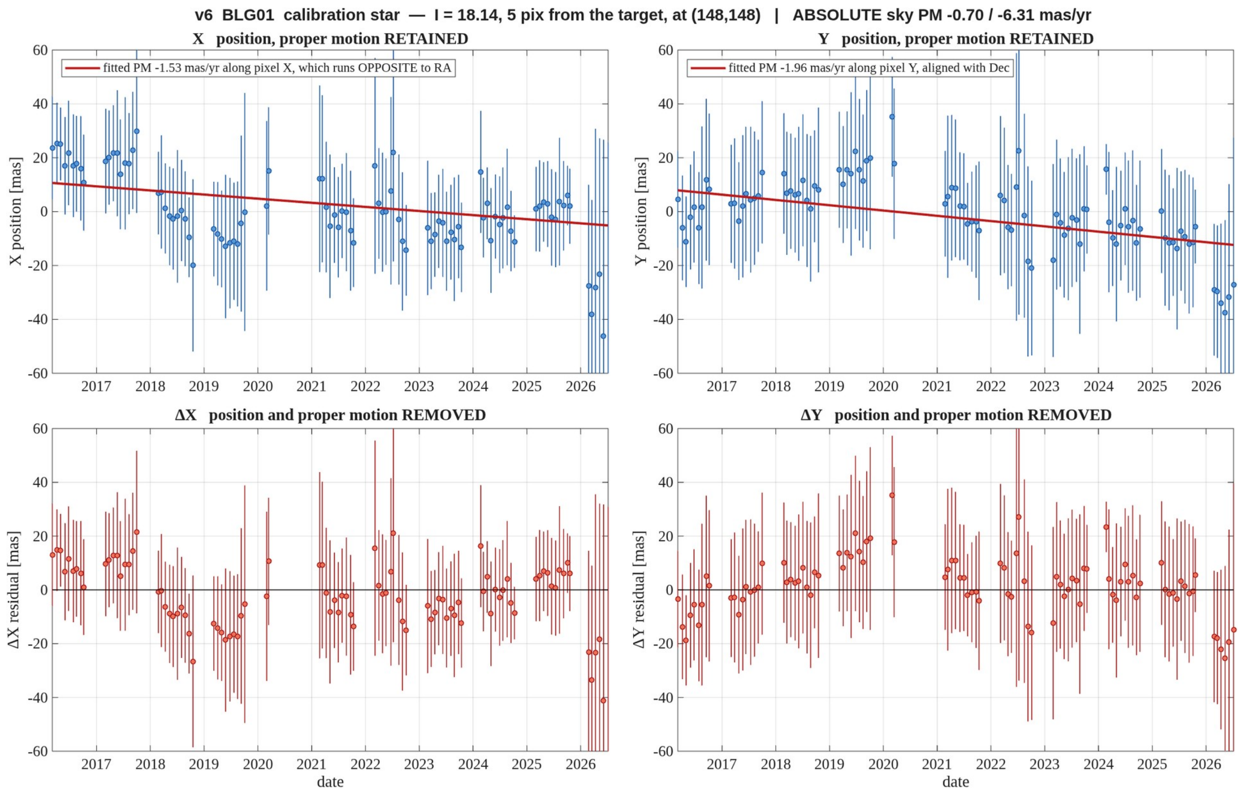


Figure 17. **BLG01** — I = 18.14, 4.5 pix away — **blended with the target**, see section 7. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG01_m18_d04.png`

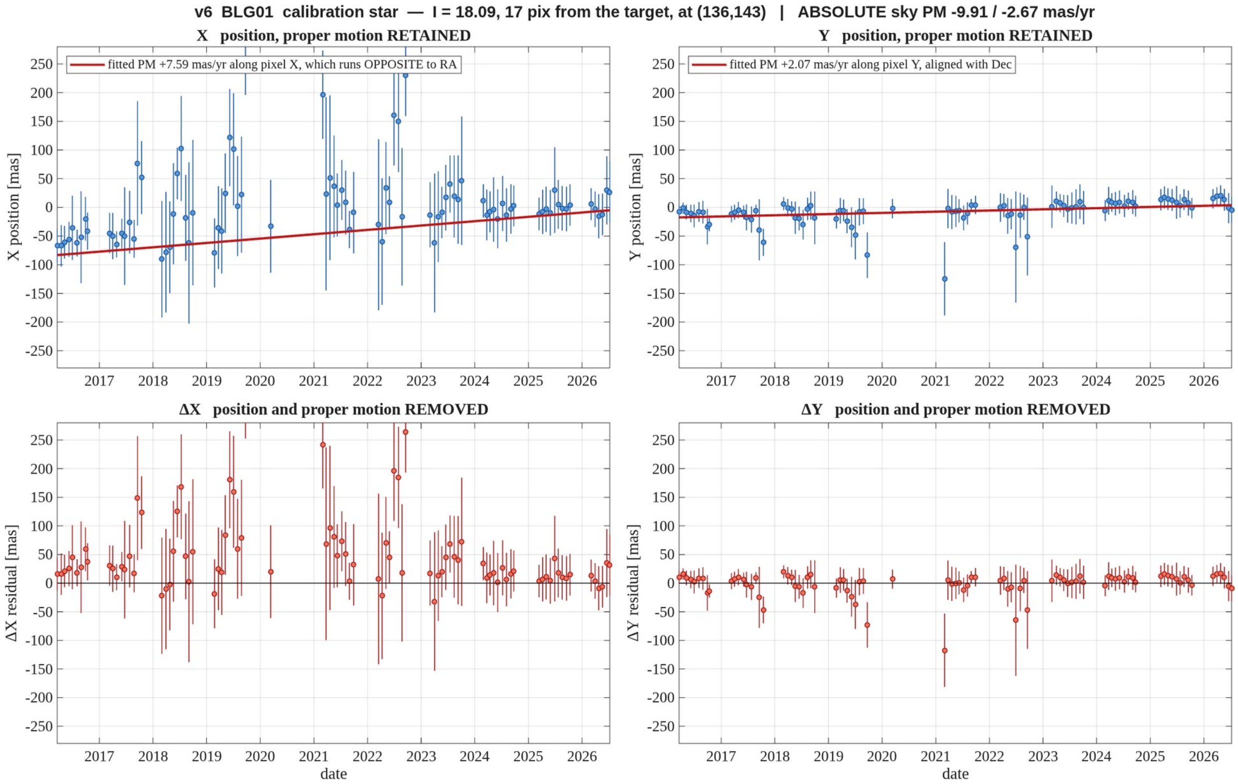


Figure 18. **BLG01** — I = 18.09, 17 pix away — poorly measured, see section 7. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG01_m18_d17.png`

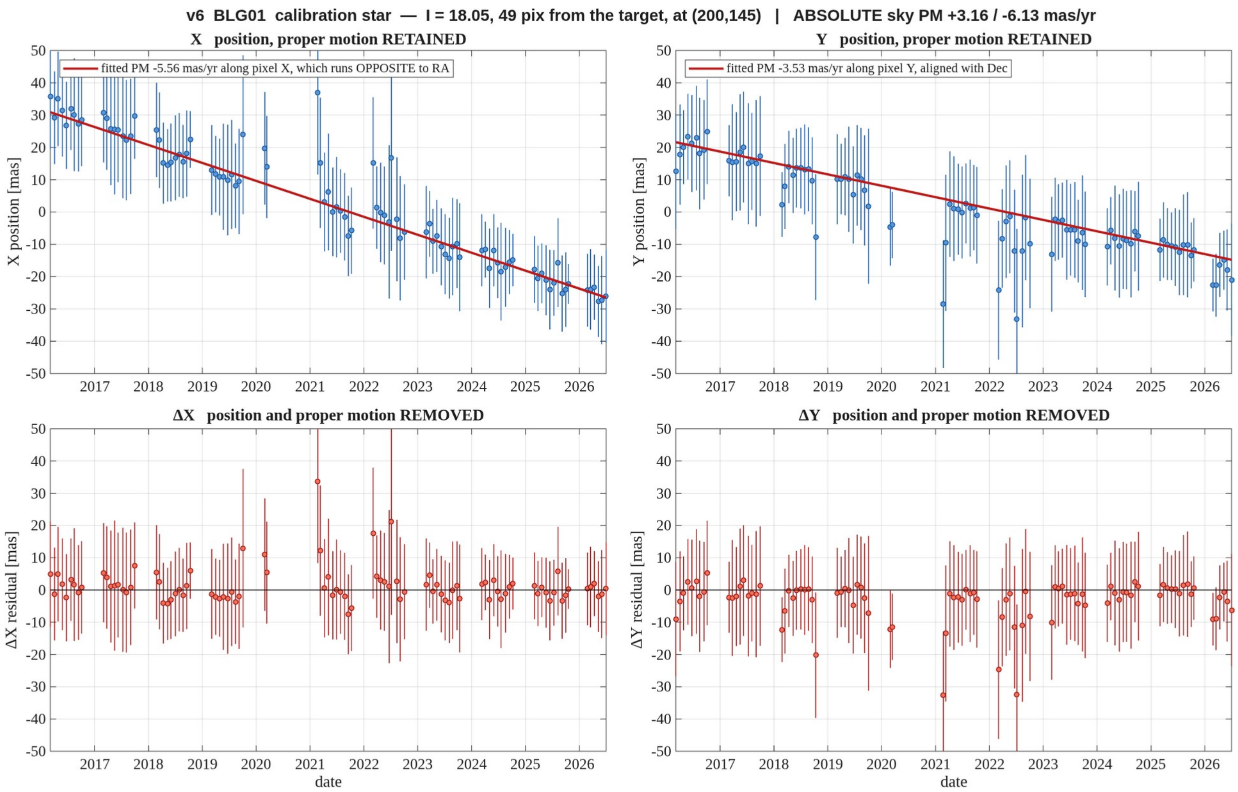


Figure 19. **BLG01** — I = 18.05, 48 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG01_m18_d48.png`

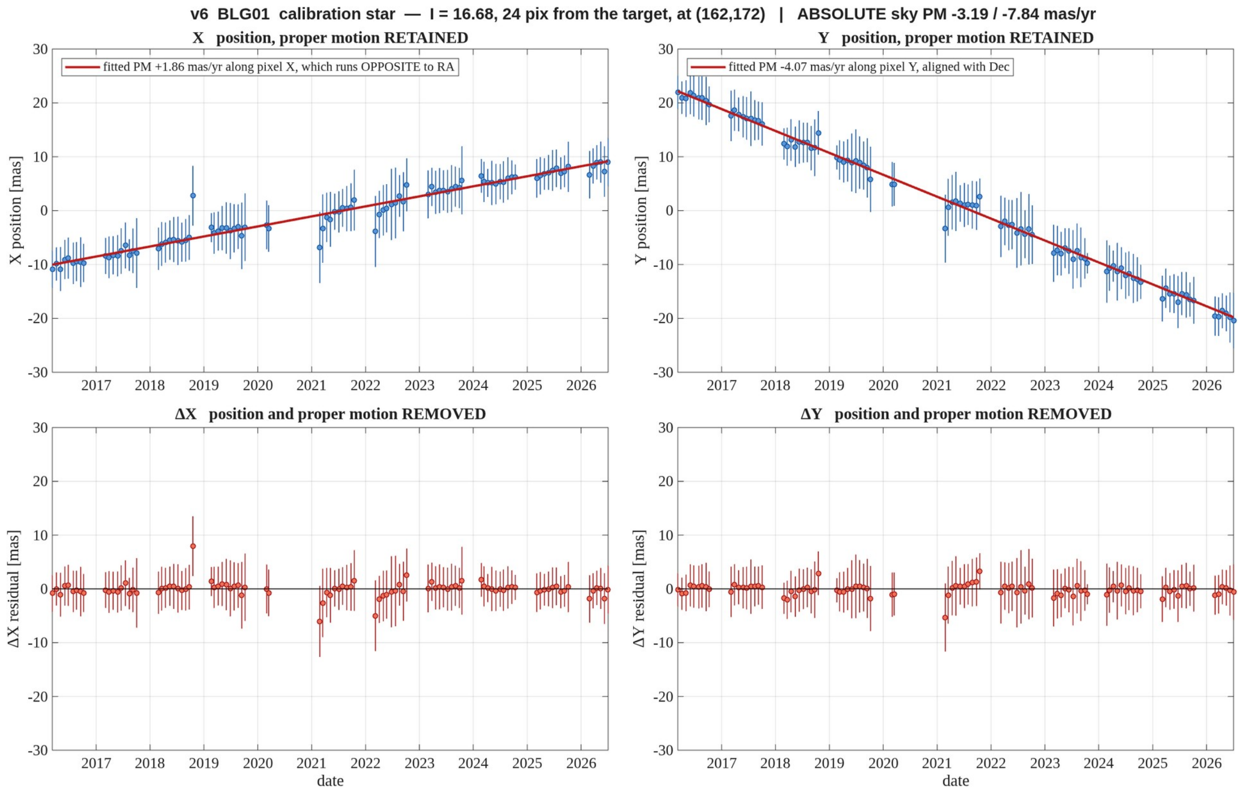


Figure 20. **BLG01** — I = 16.68, 24 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG01_m16_d24.png`

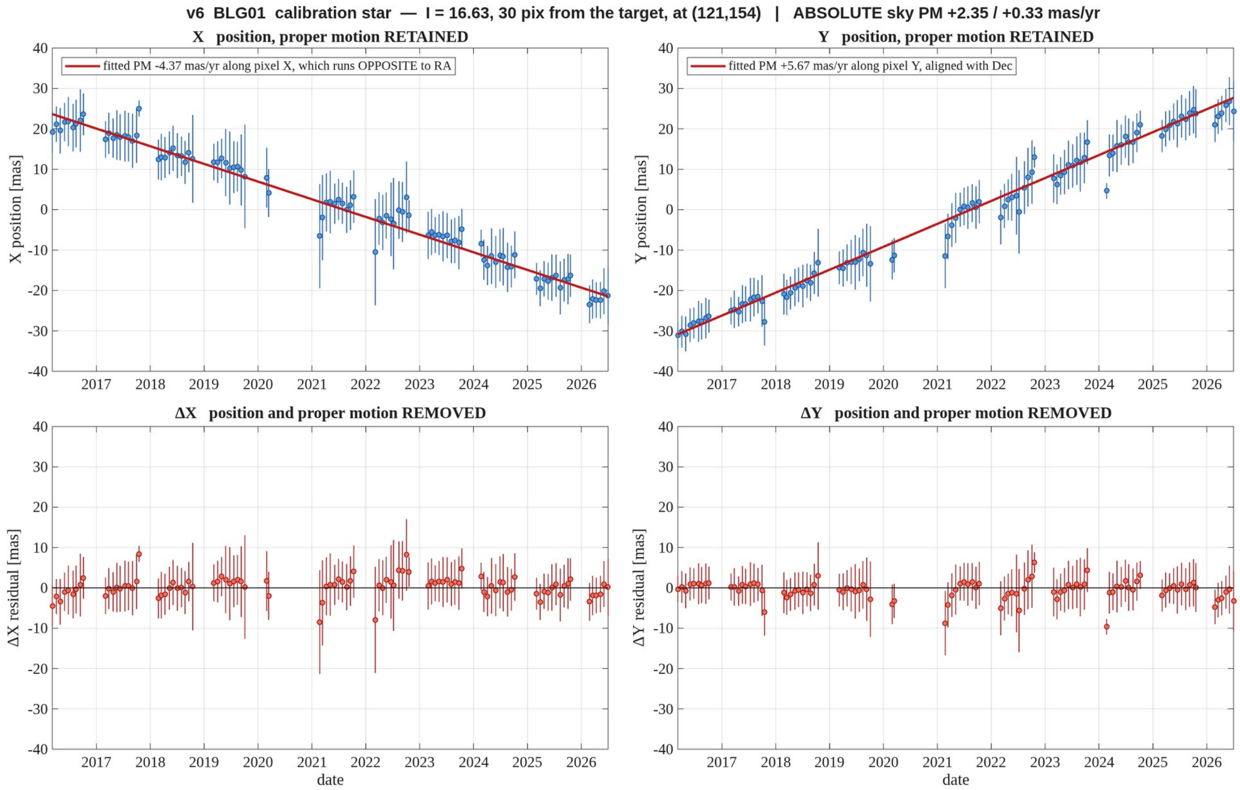


Figure 21. **BLG01** — I = 16.63, 30 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG01_m16_d30.png`

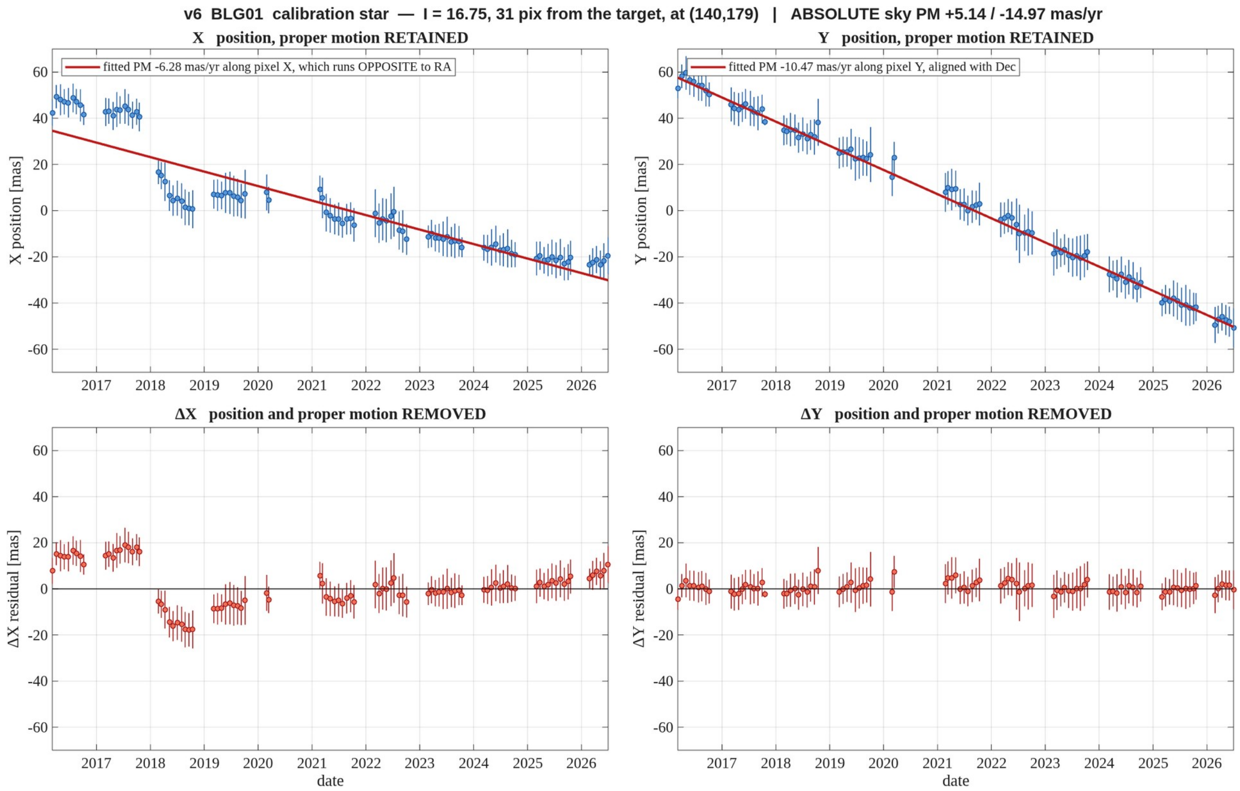


Figure 22. **BLG01** — I = 16.75, 31 pix away. Binned in one sidereal month, error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the absolute sky motion with the full gauge removed; the legend gives it along the pixel axis, and **pixel X runs opposite to RA**, so the two carry opposite signs in X.

`report_v6_motion_BLG01_m16_d31.png`